



Multicomponent, Multiphase Thermodynamics of Swelling Porous Media with Electroquasistatics: I. Macroscale Field Equations

LYNN SCHREYER BENNETHUM^{1,*} and JOHN H. CUSHMAN^{2,**}

¹*Center for Computational Mathematics, University of Colorado at Denver, Campus Box 170, P.O. Box 173364, Denver, CO 80217-3364, U.S.A. e-mail: bennethum@math.cudenver.edu.*

²*Center for Applied Math, Math Sciences Building, Purdue University, W. Lafayette, IN 47907, U.S.A.*

Abstract. A systematic development of the macroscopic field equations (conservation of mass, linear and angular momentum, energy, and Maxwell's equations) for a multiphase, multicomponent medium is presented. It is assumed that speeds involved are much slower than the speed of light and that the magnitude of the electric field significantly dominates over the magnetic field so that the electroquasistatic form of Maxwell's equations applies. A mixture formulation for each phase is averaged to obtain the macroscopic formulation. Species electric fields are considered, however it is assumed that it is the total electric field which contributes to the electrically induced forces and energy. The relationships between species and bulk phase variables and the macroscopic and microscopic variables are given explicitly. The resulting field equations are of relevance to many practical applications including, but not limited to, swelling clays (smectites), biopolymers, biological membranes, pulsed electrophoresis, and chromatography.

Key words: mixture theory, electrodynamics, averaging, swelling.

Nomenclature

Superscripts, Subscripts, and Other Notations

$\cdot^{\alpha j}$	j th component of α -phase on mesoscale.
$\cdot^{\alpha}$	α -phase on mesoscale.
$\cdot^{\sim}$	fluctuation from averaged quantity.
$\cdot^{\wedge}$	denotes exchange from other interface or phase.
$\cdot^{k,l}$	difference of the two quantities, that is, $\cdot^k - \cdot^l$.
$\cdot _{\alpha j}$	microscopic property of constituent j in phase [subscript] (non-averaged).

Latin Symbols

$\delta A_{\alpha\beta}$	portion of $\alpha\beta$ -interface in representative elementary volume (REV).
$b^{\alpha j}, b^{\alpha}$	external entropy source [J/(kg-s-°K)].
$C^{\alpha j}$	mass fraction of j th component [-].
$D^{\alpha j}, D^{\alpha}$	electric displacement [C/m ²].

*Author for correspondence

**Present address: Department of Molecular and Cellular Biology, Harvard University, 16 Divinity Ave. Cambridge, MA 01238, U.S.A.

$\widehat{d}^{\alpha j}$	net effect other constituents have on the charge of constituent j within phase α in Gauss' law [C/m ³].
$\widehat{d}_\beta^{\alpha j}$	net effect phase β has on the charge of constituent j in phase α in Gauss' law [C/m ³].
$e^{\alpha j}, e^\alpha$	energy density [J/kg].
$\widehat{e}_\beta^{\alpha j}$	rate of mass transfer from phase [subscript] to phase [superscript] per unit mass density [1/s].
$E^{\alpha j}, E^\alpha$	electric field intensity generated by (the j th constituent of) phase α [V/m].
$g^{\alpha j}, g^\alpha$	external supply of momentum (gravity) [m/s ²].
$g_I^{\alpha j}, g_I^\alpha$	internal supply of momentum due to fluctuations in Kelvin and Lorentz forces [m/s ²].
$h^{\alpha j}, h^\alpha$	external supply of energy [J/(kg-s)].
$\widehat{h}_\beta^{\alpha j}, \widehat{h}^\alpha$	free current density induces by other species within the same phase within Ampère's law [A/m ²].
$\widehat{h}_\beta^{\alpha j}, \widehat{h}_\beta^\alpha$	free current density induces by other phases on species j in Ampère's law [A/m ²].
$\widehat{i}^{\alpha j}$	rate of momentum gain due to interaction with other species within the same phase per unit mass density [N/kg].
$H^{\alpha j}, H^\alpha$	magnetic field intensity [A/m].
$\widehat{i}^{\alpha j}$	rate of momentum gain due to interaction with other species within the same phase per unit mass density [N/kg].
$J^{\alpha j}, J^\alpha$	free current density in a fixed frame of reference [A/m ²].
$\mathcal{J}^{\alpha j}, \mathcal{J}^\alpha$	free current density in the material frame of reference [A/m ²] = [C/m ² -s]; $\mathcal{J}^{\alpha j} = J^{\alpha j} - q_e^{\alpha j} \mathbf{v}^{\alpha j}$.
$\widehat{m}^{\alpha j}$	rate of angular momentum gain due to interaction with other species within the same phase per unit mass density [N-m/kg].
$\widehat{m}_\beta^{\alpha j}$	rate of angular momentum gain by constituent j in phase α due to interaction with phase β [N-m/kg].
$M^{\alpha j}$	rate of angular momentum gain due to the microscale angular momentum terms, see Appendix C [N-m/kg].
n^α	unit normal vector pointing out of α -phase within mesoscopic REV [-].
$P^{\alpha j}, P^\alpha$	polarization density averaged over α -phase [C/m ²].
$q_e^{\alpha j}, q_e^\alpha$	charge density averaged over α -phase [C/m ³].
$\widehat{m}^{\alpha j}$	rate of gain of angular momentum of constituent j from other constituents in phase α [m ² /s ²].
$q^{\alpha j}$	partial heat flux vector for the j th component of phase [J/(m ² -s)].
q^α	heat flux vector for phase α [J/(m ² -s)].
$\widehat{q}^{\alpha j}$	net rate of charge density gain by species j due to interaction with other species within phase α (does not include that gained to due mass transfer) [C/m ³ -s].
$\widehat{Q}^{\alpha j}$	rate of energy gain due to interaction with other species within the same phase per unit mass density not due to mass or momentum transfer [J/(kg-s)].
$\widehat{Q}_\beta^{\alpha j}, \widehat{Q}_\beta^\alpha$	energy transfer rate from phase [subscript] to phase [superscript] per unit mass density not due to mass or momentum transfer [J/(kg-s)].
$\mathbf{r}$	microscale spatial variable [m].
$\widehat{\tau}^{\alpha j}$	rate of mass gain due to interaction with other species within the same phase per unit mass density [1/s].
t	time [s].
T	temperature [°K].
$\mathbf{t}^{\alpha j}$	partial stress tensor for the j th component for phase [N/m ²].
$\mathbf{t}^\alpha$	total stress tensor for the phase [N/m ²].

$\hat{T}_{\beta}^{\alpha j}, \hat{T}_{\beta}^{\alpha}$	rate of momentum transfer through mechanical interactions from phase [subscript] to phase [superscript] per unit mass density [N/kg].
$\mathbf{v}^{\alpha j}, \mathbf{v}^{\alpha}$	velocity [m/s].
δV	representative elementary volume (REV).
δV_{α}	portion of α -phase in REV.
$\mathbf{w}_{\alpha\beta}^j$	velocity of constituent j at interface between phases α and β [m/s].
$\mathbf{x}$	macroscale spatial variable [m].
$z^{\alpha j}$	charge per unit mass density of constituent j in phase α [C/kg], $q_e^{\alpha j} = \rho^{\alpha j} z^{\alpha j}$.
$\hat{Z}_{\beta}^{\alpha j}, \hat{Z}_{\beta}^{\alpha}$	rate of exchange of charge of constituent j from phase β to phase α per unit mass [C/kg-s].

Greek Symbols

γ^{α}	indicator function which is 1 if in mesoscopic region α and zero otherwise.
ε^{α}	volume fraction of α -phase in mesoscale REV [-].
ε_0	permittivity in a vacuum. In MKS units, the permittivity has the value of 8.854×10^{-12} farads per meter.
ϵ_{klm}	permutation tensor.
$\hat{\Lambda}^{\alpha j}, \hat{\Lambda}^{\alpha}$	entropy production per unit mass density [J/(kg-s-°K)].
ξ	microscale spatial variable which varies over REV for fixed $\mathbf{x}$: $\mathbf{r} = \mathbf{x} + \xi$ [m].
$\eta^{\alpha j}, \eta^{\alpha}$	entropy [J/(kg-°K)].
$\hat{\eta}^{\alpha j}$	entropy gain due to interaction with other species within the same phase/interface per unit mass density [J/(kg-s-°K)].
$\phi^{\alpha j}$	partial entropy flux vector for the j th component for phase [J/(m ² -s-°K)].
ϕ^{α}	total entropy flux vector for the phase [J/(m ² -s-°K)].
$\hat{\Phi}_{\beta}^{\alpha j}, \hat{\Phi}_{\beta}^{\alpha}$	entropy transfer through mechanical interactions from phase [subscript] to phase [superscript] per unit mass [J/(kg-s-°K)].
$\rho^{\alpha j}$	partial mass density of j th component of α -phase [kg/m ³] so that $\varepsilon^{\alpha} \rho^{\alpha j}$ is the total mass of j th constituent in phase α divided by the volume of REV.
ρ^{α}	mass density of α -phase averaged over α -phase [kg/m ³].
$\hat{\sigma}^{\alpha j}$	induced curl of electric field of species j due to presence of other species within the same phase. See Faraday's law [V/m].
$\hat{\sigma}_{\beta}^{\alpha j}$	effect constituent j of phase β has on the curl of the electric field of phase α [V/m].

1. Introduction

We attempt to address the following issue: in a composite or porous medium, how do the electrothermodynamic variables of each individual constituent contribute to the electrothermodynamic variables of the mixture as a whole? In other words, given information about individual materials, determine the relationship between the electrothermodynamic properties of individual components and the electrothermodynamic properties of the averaged multiphase multicomponent material. We combine mixture theory and averaging to obtain macroscopic field equations for a deformable porous medium in which species may induce different electric fields but forces on the species are determined by the net electric field of the mixture.

The constituent forms of Maxwell's equations were first developed by Kelly (1964), although he did not consider deformable media. A binary mixture of electromagnetic fluids has been used to model plasma with single electric and magnetic

fields, (see e.g. Kulsrud, 1983). Benach and Müller (1973) applied single electric and magnetic fields to model a mixture of dielectric fluids. The idea of using multiple electric fields has been used historically in modeling a binary mixture of superconducting fluids (Coffey, 1992). Eringen (1998) developed a mixture theoretic approach for a multiple constituent electromagnetic deformable medium with multiple electromagnetic fields and applied the model to combinations of conducting/superconducting fluids and elastic solids. This article differs from Eringen (1998) in that multiple phases are considered, a different philosophy is taken in the electric fields of constituents accounting for energy and momentum balance, and the mixture may swell. Mixture theory alone does not allow for this development. An upscaling approach is required to map microscopic information into well-defined macroscopic variables.

On the microscale one can distinguish between phases or materials, while the macroscale is defined to be the scale at which the material appears to be homogeneous. There are several upscaling techniques to choose between: homogenization (matched asymptotics) (Sanchez-Palencia, 1980; Douglas and Arbogast, 1990; Bensoussan *et al.*, 1999), volume averaging in the sense of Whitaker (1967, 1969, 1990, 2000a, b), spectral integral methods (Gelhar and Axness, 1983), generalized Taylor–Aris methods (Gelhar and Axness, 1983), and hybrid mixture theory (Hassanizadeh and Gray, 1979a, b; Achanta *et al.*, 1994; Bennethum and Cushman, 1996). All but the latter of these methods upscale field equations and constitutive equations from the microscale to the macroscale. The hybrid mixture theoretic approach that we adapt does not upscale constitutive relations.

At the macroscale, the medium is viewed as a continuum, where thermodynamic properties for each constituent of each phase exist spatially everywhere. The relationships between the microscale variables and the macroscale variables are explicitly given. For ease of exposition, we assume that interfacial properties such as excess mass density, free charge on interfaces, and interface currents, are negligible; although the present theory can be extended to incorporate these effects (Gray and Hassanizadeh, 1991). The medium is referred to as multiphase, but it is understood that this includes composites (e.g. a medium composed of two solid materials), or a porous medium, where the multiphases may be gas, immiscible liquids, and/or solid.

Maxwell's equations involve species electric fields, but the momentum and energy equations for species are based on the philosophy that the force or work induced by the electric field acting on a species is generated by the *total* electric field. This differs from what has been done in the past (Kelly, 1964; Eringen, 1998). The species electric fields are defined through Gauss' law since the charge and polarization density are well defined for a species. The species electric fields then manifest themselves in the macroscale equations through terms involving gradients or time-rate-of-change of the volume fraction.

The resulting field equations may be used to obtain restrictions on admissible constitutive relations by applying the methods of either extended thermodynamics

(Müller and Ruggeri, 1993), or rational thermodynamics (Coleman and Noll, 1963; Eringen, 1967). The need for these equations, and the explicit relationships between the macroscopic and microscopic variables is demonstrated by the increasingly complex porous media being studied. Recent examples include Huyghe and Janssen (1999) and Gu *et al.* (1999), who use a simplified version of the equations developed herein to model an incompressible porous medium composed of an electrically charged solid phase saturated with an ionic fluid.

2. Basic Setting

In this section the philosophy for obtaining the mixture theoretic equations is provided. These equations include mass conservation, balance of linear and angular momenta, conservation of energy, entropy balance, and Maxwell's equations. The complete form of these equations for a bulk phase (single phase with no species) are given by Eringen (1990) and Tiersten (1990), and we adapt their formulation and notation.

It is assumed that the dominant field source is the electric field and that velocities are small compared to the speed of light (nonrelativistic). Following Melcher (1981), the electroquasistatic system of equations are obtained by nondimensionalizing Maxwell's equations, expanding each variable in a Taylor series about the variable representing the ratio of the electromagnetic wave transit time to the characteristic time of the problem, and taking the zero-order equations as the quasistatic formulation. The difference between the electroquasistatic formulation and the magnetoquasistatic formulation arises in the choice of the normalizing parameters: in the former a reference electric field is used, and in the latter a reference magnetic field is used.

It is necessary to postulate governing field equations which hold for each specie. Following Truesdell (1984), we adhere to the following principles: (1) all properties of the mixture must be mathematical consequences of properties of the constituents; (2) to describe the motion of a constituent, we may perceive it as being isolated from the rest of the mixture, provided we allow properly for the actions of the other constituents; (3) the motion of the mixture is governed by the same equations as is a single body. Further, due to the incorporation of electric fields, some assumptions must be made regarding the form of the momentum and energy balance laws. It is assumed that (4) the 'primitive' form of the balance laws is the one incorporating the work and force, and not, for example, the electrostress tensors. This is in contrast to Landau and Lifschitz (1990) and Castellanos (1998) in which the electromagnetic stresses are considered the primitive quantity. The forms are equivalent up to the classical Maxwell's equations, but are not equivalent in the mixture formulation. Until experiments prove otherwise, the choice is purely philosophical. The last principle is that (5) within the electrical work and force terms in the energy and momentum balance equations, respectively, it is the *total* electric field which acts on the specie, and not just the electric field associated with

the species. This last assumption guarantees that principles (1), (2), and (3) are not violated. For the species equations for a single phase, see for example Eringen (1998) and Benethum and Cushman (1999).

The microscale species equations are averaged to the macroscale. At this new scale, each thermodynamic variable is defined spatially everywhere, so that if there are three phases, with each phase containing N constituents, the medium is viewed as $3N$ overlaying continua. The relationships between the microscopic and macroscopic variables are given in Appendix C. The relationships between the species' variables and the bulk phase variables are obtained by summing the field equation over all constituents. These latter relationships are given in Appendix B.

3. Averaging Procedure

Consider a multiconstituent multiphase medium where a phase is denoted by Greek letters (α, β, γ), and a constituent or species is denoted by j , $j = 1, \dots, N$. For ease of exposition we assume that interfaces contain no thermodynamic properties. Consequently, it is assumed no amount of mass, momentum, energy, charge, current, etc. is lost when being transferred between phases. Interfacial effects can be included by pursuing any of the approaches of Moeckel (1975), Gray and Hassanizadeh (1989, 1991).

Assuming no internal surface discontinuities, the constituent, microscopic field equations of mass, charge, linear momentum, angular momentum, energy, and entropy can be expressed for a given phase, α , as (following the notation of Eringen (1967)):

$$\frac{\partial}{\partial t}(\rho^j \Psi^j) + \nabla \cdot (\rho^j \mathbf{v}^j \Psi^j) - \nabla \cdot \mathbf{i}^j - \rho^j f^j - F^j = \rho^j G^j + \rho^j \hat{\Psi}^j, \quad (1)$$

where Ψ^j is the mass-average (over the phase) thermodynamic property of constituent j , $\mathbf{v}^j$ is the mass-average velocity vector, ρ^j is the mass density, $\mathbf{i}^j$ is the flux vector, f^j is the body source, F^j is the supply due to electrical effects, G^j is the net production, and $\hat{\Psi}^j$ represents the influx of Ψ from all other constituents (e.g. due to chemical reactions). If there is only one constituent, $\hat{\Psi}^j$ is zero. For each of the respective equations, the quantities given in Table I are used. For conciseness, we introduce the force, couple, and work due to the electric fields, respectively:

$$\mathbf{F}_e^j = q_e^j \mathbf{E} + \mathbf{P}^j \cdot \nabla \mathbf{E}, \quad (2)$$

$$\begin{aligned} &= -\nabla \cdot \left(\mathbf{D}^j \mathbf{E} - \frac{1}{2} \varepsilon_o \mathbf{E}^j \cdot \mathbf{E} \mathbf{I} \right) + \hat{d}^j \mathbf{E} + \\ &\quad + \frac{1}{2} \varepsilon_o (\nabla \mathbf{E} \cdot \mathbf{E}^j - \nabla \mathbf{E}^j \cdot \mathbf{E}), \end{aligned} \quad (3)$$

$$\mathbf{C}_e^j = \mathbf{P}^j \times \mathbf{E}, \quad (4)$$

Table I. Quantities for Equation (1) [15]

Quantity	Ψ	$\mathbf{i}$	f	F	$\hat{\Psi}$	G
Mass	1	$\mathbf{0}$	0	0	$\hat{r}$	0
Charge	z	$-\mathcal{J}$	0	0	$\hat{q}/\rho + z\hat{r}$	0
Linear momentum	$\mathbf{v}$	$\mathbf{t}$	$\mathbf{g} + \mathbf{g}_I$	$\mathbf{F}_e$	$\hat{\mathbf{i}} + \hat{r}\mathbf{v}$	0
Angular momentum	$\mathbf{r} \times \mathbf{v}$	$\mathbf{r} \times \mathbf{t}$	$\mathbf{r} \times \mathbf{g}$	$\mathbf{r} \times \mathbf{F}_e + \mathbf{C}_e$	$\mathbf{r} \times (\hat{\mathbf{i}} + \hat{r}\mathbf{v}) + \hat{m}$	0
Energy	$e + \frac{1}{2}v^2$	$\mathbf{t} \cdot \mathbf{v} + \mathbf{q}$	$\mathbf{g} \cdot \mathbf{v} + h$	W_e	$\hat{Q} + \hat{\mathbf{i}} \cdot \mathbf{v} + \hat{r}(e + \frac{1}{2}v^2)$	0
Entropy	η	ϕ	b	0	$\hat{\eta} + \hat{r}\mathbf{v}$	Λ

$$W_e^j = \mathbf{J}^j \cdot \mathbf{E} + \frac{\partial \mathbf{P}^j}{\partial t} \cdot \mathbf{E} + \nabla \cdot (\mathbf{v}^j \mathbf{E} \cdot \mathbf{P}^j), \quad (5)$$

$$= \frac{\partial}{\partial t} \left(-\frac{1}{2} \varepsilon_0 \mathbf{E}^j \cdot \mathbf{E} \right) - \nabla \cdot (\mathbf{E} \times \mathbf{H}^j) + \nabla \cdot (\mathbf{P}^j \mathbf{v}^j \cdot \mathbf{E}) + \\ + \hat{\mathbf{h}}^j \cdot \mathbf{E} - \frac{1}{2} \varepsilon_o \left(\frac{\partial \mathbf{E}^j}{\partial t} \cdot \mathbf{E} - \frac{\partial \mathbf{E}}{\partial t} \cdot \mathbf{E}^j \right), \quad (6)$$

where $\mathbf{E}^j$ is the species electric field (discussed below), and $\mathbf{E}$ is the total electric field, obtained by summing all of the species electric fields. Definitions of all terms are given in Appendix A. The forms of $\mathbf{F}_e^j$ and W_e^j are a consequence of which form one assumes to be the most primitive. If one assumes that the second forms of $\mathbf{F}_e^j$ and W_e^j , (3, 6), are the most primitive then miscellaneous terms would appear in the first forms.

The averaging procedure is based on ideas laid down in Eringen and Suhubi (1964), Slattery (1967), Whitaker (1967, 1969). Several methods are available, but we choose the computationally simplest. Equations are averaged by weighted integration using the indicator function of the α -phase. To avoid the mathematical difficulties of, for example, defining a derivative of the averaged quantities resulting from using such a weighting function, one must treat the averaged quantity as a distribution (Schwartz, 1950; Richards and YOUN, 1990). It should be noted that using this simple weight function may mean that the averaged value may not represent the actual values being measured. To account for the measuring technique, one needs to choose a weight function, which represents the instrument used to measure the physical properties (Cushman, 1984). Extensions of the present theory to such cases are straight forward.

Let δV be a volume, δV_α the portion of δV in the α -phase, and $\delta A_{\alpha\beta}$ the portion of the interface within δV . It is assumed that δV_α and $\delta A_{\alpha\beta}$ are isolated simply connected regions. If the magnitude of δV is denoted by $|\delta V|$ then the volume fraction can be expressed as

$$\varepsilon^\alpha(\mathbf{x}, t) = \frac{|\delta V_\alpha|}{|\delta V|}, \quad (7)$$

so that

$$\sum_\alpha \varepsilon^\alpha = 1. \quad (8)$$

The indicator function is

$$\gamma_\alpha(\mathbf{r}, t) = \begin{cases} 1 & \text{if } \mathbf{r} \in \delta V_\alpha, \\ 0 & \text{if } \mathbf{r} \in \delta V_\beta, \quad \beta \neq \alpha. \end{cases}$$

We may write $\mathbf{r} = \mathbf{x} + \boldsymbol{\xi}$, where $\mathbf{x}$ is the macroscale spatial variable, and $\boldsymbol{\xi}$ varies over δV . To obtain the macroscale equations formally, one multiplies Equation (1) by $\gamma_\alpha(\mathbf{r})$, integrates over δV with respect to $\boldsymbol{\xi}$ ($\mathbf{x}$ is held fixed) and divides by $|\delta V|$. Then in order to obtain equations of the forms which mirror the microscale equations, the following theorem is applied to interchange the order of differentiation and integration.

THEOREM 1. *If $\mathbf{w}_{\alpha\beta}$ is the microscopic velocity of interface $\alpha\beta$ and $\mathbf{n}_\alpha$ is the outward unit normal vector of δV_α indicating the integrand should be evaluated in the limit as the $\alpha\beta$ -interface is approached from the α -side then*

$$\begin{aligned} \frac{1}{|\delta V|} \int_{\delta V} \frac{\partial f}{\partial t} \gamma_\alpha d\mathbf{v}(\boldsymbol{\xi}) &= \frac{\partial}{\partial t} \left[\frac{1}{|\delta V|} \int_{\delta V} f \gamma_\alpha d\mathbf{v}(\boldsymbol{\xi}) \right] - \\ &\quad - \sum_{\beta \neq \alpha} \frac{1}{|\delta V|} \int_{\delta A_{\alpha\beta}} f \mathbf{w}_{\alpha\beta} \cdot \mathbf{n}_\alpha d\mathbf{a}(\boldsymbol{\xi}), \end{aligned} \quad (9)$$

$$\begin{aligned} \frac{1}{|\delta V|} \int_{\delta V} \nabla_{\mathbf{r}} f \gamma_\alpha d\mathbf{v}(\boldsymbol{\xi}) &= \nabla_{\mathbf{x}} \left[\frac{1}{|\delta V|} \int_{\delta V} f \gamma_\alpha d\mathbf{v}(\boldsymbol{\xi}) \right] + \\ &\quad + \sum_{\beta \neq \alpha} \frac{1}{|\delta V|} \int_{\delta A_{\alpha\beta}} f \mathbf{n}_\alpha d\mathbf{a}(\boldsymbol{\xi}). \end{aligned} \quad (10)$$

As an example, consider the conservation of charge

$$\nabla \cdot (\mathcal{J}^j + q_e^j \mathbf{v}^j) + \frac{\partial q_e^j}{\partial t} = \hat{q}^j + \rho^j z^j \hat{r}^j. \quad (11)$$

We obtain the macroscale equation by formally multiplying this equation by γ_α , integrating over δV , and then dividing by $|\delta V|$. Using the theorem, we have for the time derivative term,

$$\begin{aligned}
& \frac{1}{|\delta V|} \int_{\delta V} \frac{\partial q_e^j}{\partial t} \gamma_\alpha \, dv \\
&= \frac{\partial}{\partial t} \left[\frac{1}{|\delta V|} \int_{\delta V} q_e^j \gamma_\alpha \, dv \right] - \sum_{\beta \neq \alpha} \frac{1}{|\delta V|} \int_{\delta A_{\alpha\beta}} q_e^j \mathbf{w}_{\alpha\beta} \cdot \mathbf{n}_\alpha \, da \\
&= \frac{\partial}{\partial t} (\varepsilon^\alpha q_e^{\alpha j}) - \sum_{\beta \neq \alpha} \frac{1}{|\delta V|} \int_{\delta A_{\alpha\beta}} q_e^j \mathbf{w}_{\alpha\beta} \cdot \mathbf{n}_\alpha \, da. \tag{12}
\end{aligned}$$

The first term remains on the left-hand side, the term involving summation over β is absorbed into the exchange term, $\widehat{Z}_\beta^\alpha$. A similar process is done to average the term $\nabla \cdot (\mathcal{J}^j + q_e^j \mathbf{v}^j)$. In averaging the exchange term becomes

$$\begin{aligned}
\frac{1}{|\delta V|} \int_{\delta V} \rho^j z^j \widehat{\mathbf{r}}^j \gamma_\alpha \, dv &= \varepsilon^\alpha \langle \rho^j z^j \widehat{\mathbf{r}}^j \rangle^\alpha = \varepsilon^\alpha \rho^{\alpha j} \overline{z^j \widehat{\mathbf{r}}^j}^\alpha \\
&= \varepsilon^\alpha \rho^{\alpha j} z^{\alpha j} \widehat{\mathbf{r}}^{\alpha j} + (\varepsilon^\alpha \rho^{\alpha j} \overline{z^j \widehat{\mathbf{r}}^j}^\alpha - \varepsilon^\alpha \rho^{\alpha j} z^{\alpha j} \widehat{\mathbf{r}}^{\alpha j}), \tag{13}
\end{aligned}$$

where the difference in the mass average of the product and the product of the mass average gets absorbed into $\widehat{q}^{\alpha j}$. Using the definitions in Appendix C, we obtain the macroscopic form of the conservation of charge as written in the next section. It should be noted that for the exchange terms between phases, we explicitly separate the effects due to mass transfer, which explains the additional term $\varepsilon^\alpha \rho^{\alpha j} z^{\alpha j} \widehat{\mathbf{e}}_\beta^{\alpha j}$. This quantity is subtracted out in the term $\widehat{Z}_\beta^\alpha$ so that

$$\widehat{Z}_\beta^\alpha = \frac{1}{\rho^{\alpha j} |\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} \left[q_e^j (\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j) - \mathcal{J}^j \right] \cdot \mathbf{n}^\alpha \, da - z^{\alpha j} \widehat{\mathbf{e}}_\beta^{\alpha j}. \tag{14}$$

After averaging Equation (1), the system is considered to be a mixture so that each component in each phase and each bulk phase have thermodynamic properties existing at each point within the macroscopic body. The macroscopic definition of each field variable in terms of its microscopic counterpart, making *no small perturbation assumptions*, is given in Appendix C.

For more details regarding this procedure for the field equations with no electric or magnetic effects (see e.g. Slattery, 1967; Whitaker, 1967; Plumb and Whitaker, 1990). For the parallel development of Maxwell's equations with constitutive assumptions, (see del Rio and Whitaker, 2000a, b).

4. Macroscopic Conservation Equations

4.1. CONSERVATION OF MASS

The macroscopic mass balance for constituent j in phase α is

$$\frac{D^{\alpha j} (\varepsilon^\alpha \rho^{\alpha j})}{Dt} + \varepsilon^\alpha \rho^{\alpha j} (\nabla \cdot \mathbf{v}^{\alpha j}) = \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^{\alpha j} \widehat{\mathbf{e}}_\beta^{\alpha j} + \varepsilon^\alpha \rho^{\alpha j} \widehat{\mathbf{r}}^{\alpha j}, \tag{15}$$

where $D^{\alpha j}/Dt$ is the material time derivative given by

$$\frac{D^{\alpha j}}{Dt} = \frac{\partial}{\partial t} + \mathbf{v}^{\alpha j} \cdot \nabla, \quad (16)$$

ε^α is the volume fraction of phase α , $\widehat{r}^{\alpha j}$ is the rate of mass transfer to species j from other species due to chemical reactions, and $\widehat{e}_\beta^{\alpha j}$ represents the net rate of mass gained by constituent j in phase α from phase β :

$$\widehat{e}_\beta^{\alpha j} = \frac{1}{\rho^{\alpha j} |\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} \rho^j (\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j) \cdot \mathbf{n}^\alpha da, \quad (17)$$

where $\mathbf{w}_{\alpha\beta}^j$ is the velocity of species j at interface $\alpha\beta$, δV_α the portion of the averaging volume, δV , in the α -phase, $\delta A_{\alpha\beta}$ the portion of the interface within δV , $\mathbf{v}^j$ is the microscopic velocity of constituent j , and $\mathbf{n}^\alpha$ is the outward unit normal vector of δV_α . All variables are defined in Appendix A.

The bulk phase counterpart is given by

$$\frac{D^\alpha(\varepsilon^\alpha \rho^\alpha)}{Dt} + \varepsilon^\alpha \rho^\alpha (\nabla \cdot \mathbf{v}^\alpha) = \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^\alpha \widehat{e}_\beta^\alpha. \quad (18)$$

The net gain of mass of the bulk phase due to chemical reactions alone must be zero, implying that

$$\sum_{j=1}^N \rho^{\alpha j} \widehat{r}^{\alpha j} = 0, \quad \forall \alpha. \quad (19)$$

Further, as the interface is assumed to be massless, we have the restriction

$$\varepsilon^\alpha \rho^{\alpha j} \widehat{e}_\beta^{\alpha j} + \varepsilon^\beta \rho^{\beta j} \widehat{e}_\alpha^{\beta j} = 0, \quad j = 1, \dots, N. \quad (20)$$

4.2. GAUSS' LAW

For this equation we introduce the electric field for species j , $\mathbf{E}^{\alpha j}$. Let $\mathbf{D}^{\alpha j}$ be the volume averaged displacement vector for the j th component only. In terms of the electric field and polarization density the species displacement vector is

$$\mathbf{D}^{\alpha j} = \varepsilon_0 \mathbf{E}^{\alpha j} + \mathbf{P}^{\alpha j}, \quad (21)$$

where ε_0 is the permittivity in a vacuum. In MKS units, the permittivity has the value of 8.854×10^{-12} farads per meter (Melcher, 1981). The macroscopic form of Gauss' law for constituent j in phase α is

$$\nabla \cdot (\varepsilon^\alpha \mathbf{D}^{\alpha j}) - \varepsilon^\alpha q_e^{\alpha j} = \varepsilon^\alpha \widehat{d}^{\alpha j} + \sum_{\beta \neq \alpha} \varepsilon^\alpha \widehat{d}_\beta^{\alpha j}, \quad (22)$$

where $q_e^{\alpha_j}$ is the charge density of constituent j , $\widehat{d}^{\alpha_j}$ is the excess charge density due to the presence of other species, and $\widehat{d}_\beta^{\alpha_j}$ represents the effect constituent j in phase β has on the charge of the same constituent in phase α

$$\widehat{d}_\beta^{\alpha_j} = -\frac{1}{|\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} \mathbf{D}^j \cdot \mathbf{n}^\alpha \, da. \quad (23)$$

Other relations are given in Appendix C.

The species' electric fields, $\mathbf{E}^{\alpha_j}$, may be interpreted in several ways but must satisfy the restriction that the total electric field, $\mathbf{E}_T$, is the sum of the species' electric fields, that is, $\sum_{j=1}^N \mathbf{E}^{\alpha_j} = \mathbf{E}_T$. If there is no external electric field then $\mathbf{E}_T$ is just the electric field generated by all the species. One natural physical interpretation of $\mathbf{E}^{\alpha_j}$ is that of the electric field generated by species j of phase α . This, however, leads to a problem in incorporating the externally applied electric field. One could treat the external electric field as the presence of another species, say species N , in which this 'external' species has no other electrothermodynamic properties, that is, species N would have no charge density, no associated polarization, etc. and these terms would be set to zero in the governing equations. Alternately, a portion of the external field could be assigned to each $\mathbf{E}^{\alpha_j}$ so that $\mathbf{E}^{\alpha_j}$ is the sum of the electric field generated by species j and a weight, w^j , times the external electric field. The weights must sum to one, and may, for example, be proportional to the amount of charge, or maybe the mass fraction of species j . A less physically intuitive way of defining $\mathbf{E}^{\alpha_j}$ is to define it so that the sum of the exchange terms in Gauss' law, $\widehat{d}^{\alpha_j} + \widehat{d}_\beta^{\alpha_j}$, is zero. This would result in simplifying computations and reducing the number of unknowns. These choices have no affect on the following derivation, as long as $\mathbf{E}^{\alpha_j}$ sum to the total electric field. However, in formulating a mathematical model it is necessary to choose a particular definition and remain consistent.

Summing over j yields the bulk phase form

$$\nabla \cdot (\varepsilon^\alpha \mathbf{D}^\alpha) - \varepsilon^\alpha q_e^\alpha = \sum_{\beta \neq \alpha} \varepsilon^\alpha \widehat{d}_\beta^\alpha. \quad (24)$$

The restrictions include

$$\sum_{j=1}^N \widehat{d}^{\alpha_j} = 0, \quad \forall \alpha, \quad (25)$$

$$\varepsilon^\alpha \widehat{d}_\beta^{\alpha_j} + \varepsilon^\beta \widehat{d}_\alpha^{\beta_j} = 0, \quad j = 1, \dots, N. \quad (26)$$

The first restriction states that the net effect the constituents have on each other within phase α must be zero, and the second restriction states that the interfacial displacement is zero, which is a consequence of assuming that the interface contains neither electrical nor thermodynamical properties.

Recall that if there is no surface charge or polarization, the jump condition for the displacement between two materials is (Eringen and Maugin, 1990)

$$\mathbf{n} \cdot [[\mathbf{D}]] = 0, \quad (27)$$

where the double brackets indicate the difference of the quantity evaluated on either side of the interface. Integrating this equation over $\delta A_{\alpha\beta}$ and dividing by $|\delta V|$ shows that Equation (26) is consistent with the classical jump condition, (27).

4.3. FARADAY'S LAW

The macroscopic form of Faraday's law for constituent j in phase α is

$$\nabla \times (\varepsilon^\alpha \mathbf{E}^{\alpha j}) = \varepsilon^\alpha \hat{\boldsymbol{\sigma}}^{\alpha j} + \sum_{\beta \neq \alpha} \varepsilon^\alpha \hat{\boldsymbol{\sigma}}_\beta^{\alpha j}, \quad (28)$$

or in indicial notation

$$(\varepsilon^\alpha E_k^{\alpha j})_{,l} - (\varepsilon^\alpha E_l^{\alpha j})_{,k} = \varepsilon_{lkm} \varepsilon^\alpha \hat{\sigma}_m^{\alpha j} + \sum_{\beta \neq \alpha} \varepsilon_{lkm} \varepsilon^\alpha (\hat{\sigma}_\beta^{\alpha j})_m, \quad (29)$$

where repeated indices denote summation, a comma denotes differentiation, and ε_{lkm} is the permutation tensor. Here $\hat{\boldsymbol{\sigma}}^{\alpha j}$ incorporates the effect of the electric fields, $\mathbf{E}^{\alpha i}$, $i \neq j$, and $\hat{\boldsymbol{\sigma}}_\beta^{\alpha j}$, given by

$$\hat{\boldsymbol{\sigma}}_\beta^{\alpha j} = -\frac{1}{|\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} \mathbf{n}^\alpha \times \mathbf{E}^j da, \quad (30)$$

represents the effect phase β has on the electric field of phase α . Upon summing, the bulk phase form of Faraday's law becomes

$$\nabla \times (\varepsilon^\alpha \mathbf{E}^\alpha) = \sum_{\beta \neq \alpha} \varepsilon^\alpha \hat{\boldsymbol{\sigma}}_\beta^\alpha, \quad (31)$$

with restrictions

$$\sum_{j=1}^N \hat{\boldsymbol{\sigma}}^{\alpha j} = 0, \quad \forall \alpha, \quad (32)$$

$$\varepsilon^\alpha \hat{\boldsymbol{\sigma}}_\beta^{\alpha j} + \varepsilon^\beta \hat{\boldsymbol{\sigma}}_\alpha^{\beta j} = 0, \quad j = 1, \dots, N. \quad (33)$$

Equation (33) corresponds precisely with the classical jump condition across interfaces (Eringen and Maugin, 1990)

$$\mathbf{n} \times [[\mathbf{E}]] = \mathbf{0}, \quad (34)$$

if (34) is integrated over the surface $\delta A_{\alpha\beta}$.

4.4. AMPÈRE'S LAW

The macroscopic form of Ampère's law for the quasi-static case for species j of phase α is given by

$$\begin{aligned} \varepsilon^\alpha \mathbf{J}^{\alpha j} = & -\frac{\partial(\varepsilon^\alpha \mathbf{D}^{\alpha j})}{\partial t} + \nabla \times (\varepsilon^\alpha \mathbf{H}^{\alpha j}) - \nabla \times (\varepsilon^\alpha \mathbf{P}^{\alpha j} \times \mathbf{v}^{\alpha j}) + \\ & + \sum_{\beta \neq \alpha} \varepsilon^\alpha \widehat{\mathbf{h}}_\beta^{\alpha j} + \varepsilon^\alpha \widehat{\mathbf{h}}^{\alpha j}, \end{aligned} \quad (35)$$

where $\mathbf{H}^{\alpha j}$ is the magnetic field intensity, $\mathbf{J}^j$ is the free current density of constituent j measured in a fixed reference frame, and

$$\widehat{\mathbf{h}}_\beta^{\alpha j} = \frac{1}{|\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} \left[\mathbf{D}^j \mathbf{w}_{\alpha\beta}^j \cdot \mathbf{n}^\alpha - (\mathbf{H}^j - \mathbf{P}^j \times \mathbf{v}^j) \times \mathbf{n}^\alpha \right] da \quad (36)$$

represents the effect of phase β on phase α . The free current density relative to constituent j of phase α , denoted by $\mathcal{J}^{\alpha j}$, is related to the free current density relative to a fixed frame of reference, $\mathbf{J}^{\alpha j}$, by

$$\mathcal{J}^{\alpha j} = \mathbf{J}^{\alpha j} - q_e^{\alpha j} \mathbf{v}^{\alpha j}. \quad (37)$$

Summing over species gives the bulk phase version of this law,

$$\varepsilon^\alpha \mathbf{J}^\alpha = -\frac{\partial(\varepsilon^\alpha \mathbf{D}^\alpha)}{\partial t} + \nabla \times \varepsilon^\alpha \mathbf{H}^\alpha - \nabla \times (\varepsilon^\alpha \mathbf{P}^\alpha \times \mathbf{v}^\alpha) + \sum_{\beta \neq \alpha} \varepsilon^\alpha \widehat{\mathbf{h}}_\beta^\alpha, \quad (38)$$

where the restrictions are

$$\sum_{j=1}^N \widehat{\mathbf{h}}^{\alpha j} = \mathbf{0}, \quad \forall \alpha \quad (39)$$

$$\varepsilon^\alpha \widehat{\mathbf{h}}_\beta^{\alpha j} + \varepsilon^\beta \widehat{\mathbf{h}}_\alpha^{\beta j} = \mathbf{0}, \quad j = 1, \dots, N. \quad (40)$$

The jump condition associated with Ampère's law is given by (Eringen and Maugin, 1990)

$$\mathbf{n} \times [[\mathbf{H} - \mathbf{w} \times \mathbf{D}]] = \mathbf{0}, \quad (41)$$

where the polarization at the surface has been neglected. Equation (36) does not correspond directly with the jump condition given in Eringen and Maugin (1990) as (41) is derived directly from the global surface area form of Ampère's law and (36) is derived from the bulk-phase form of the law. This causes a loss of information; specifically, (36) does not include a term corresponding to $\mathbf{w}^j (\mathbf{n} \cdot \mathbf{D}^j)$.

4.5. CONSERVATION OF ELECTRIC CHARGE

This equation can be derived by taking the divergence of Ampère's law and the time derivative of Gauss' law and summing the results. Of all Maxwell's equations, this equation is the most accepted in mixture form as each component has a well-defined physical interpretation.

The conservation of charge equation at the macroscale becomes

$$\begin{aligned} \nabla \cdot (\varepsilon^\alpha \mathcal{J}^{\alpha j} + \varepsilon^\alpha q_e^{\alpha j} \mathbf{v}^{\alpha j}) + \frac{\partial}{\partial t} (\varepsilon^\alpha q_e^{\alpha j}) \\ = \varepsilon^\alpha \hat{q}^{\alpha j} + \varepsilon^\alpha \rho^{\alpha j} z^{\alpha j} \hat{r}^{\alpha j} + \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^{\alpha j} (\hat{Z}_\beta^{\alpha j} + z^{\alpha j} \hat{e}_\beta^{\alpha j}), \end{aligned} \quad (42)$$

where $z^{\alpha j}$ is the charge per unit mass and $\hat{q}^{\alpha j}$ is the rate of gain of charge density due to the presence of other constituents but not due to chemical reactions.

$$\begin{aligned} \hat{Z}_\beta^{\alpha j} = \frac{1}{\rho^{\alpha j} |\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} [q_e^j (\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j) - \mathcal{J}^j] \cdot \mathbf{n}^\alpha \, da - \\ - \frac{z^{\alpha j}}{\rho^{\alpha j} |\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} [\rho^j (\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j)] \cdot \mathbf{n}^\alpha \, da, \end{aligned} \quad (43)$$

which represents the rate of exchange of charge of constituent j from phase β to phase α not due to mass exchange. Using the continuity equation, (15), Equation (42) may be re-written as

$$\varepsilon^\alpha \rho^{\alpha j} \frac{D^{\alpha j} z^{\alpha j}}{Dt} + \nabla \cdot (\varepsilon^\alpha \mathcal{J}^{\alpha j}) = \varepsilon^\alpha \hat{q}^{\alpha j} + \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^{\alpha j} \hat{Z}_\beta^{\alpha j}. \quad (44)$$

Summing over constituents yields

$$\nabla \cdot (\varepsilon^\alpha \mathcal{J}^\alpha + \varepsilon^\alpha q_e^\alpha \mathbf{v}^\alpha) + \frac{\partial}{\partial t} (\varepsilon^\alpha q_e^\alpha) = \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^\alpha (\hat{Z}_\beta^\alpha + z^\alpha \hat{e}_\beta^\alpha), \quad (45)$$

where the following restrictions apply

$$\sum_{j=1}^N [\hat{q}^{\alpha j} + \rho^{\alpha j} z^{\alpha j} \hat{r}^{\alpha j}] = 0, \quad \forall \alpha, \quad (46)$$

$$\varepsilon^\alpha \rho^{\alpha j} (\hat{Z}_\beta^{\alpha j} + z^{\alpha j} \hat{e}_\beta^{\alpha j}) + \varepsilon^\beta \rho^{\beta j} (\hat{Z}_\alpha^{\beta j} + z^{\beta j} \hat{e}_\alpha^{\beta j}) = 0, \quad j = 1, \dots, N. \quad (47)$$

Because of the coupling between Gauss' law, Ampère's law, and the conservation of charge, there is a coupling between the exchange terms

$$\begin{aligned} \nabla \cdot \left(\varepsilon^\alpha \hat{\mathbf{h}}^{\alpha j} + \sum_{\beta \neq \alpha} \varepsilon^\alpha \hat{\mathbf{h}}_\beta^{\alpha j} \right) - \frac{\partial (\varepsilon^\alpha \hat{d}^{\alpha j})}{\partial t} - \sum_{\beta \neq \alpha} \frac{\partial (\varepsilon^\alpha \hat{d}_\beta^{\alpha j})}{\partial t} \\ = \varepsilon^\alpha \hat{q}^{\alpha j} + \varepsilon^\alpha q_e^{\alpha j} \hat{r}^{\alpha j} + \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^{\alpha j} (\hat{Z}_\beta^{\alpha j} + z^{\alpha j} \hat{e}_\beta^{\alpha j}). \end{aligned}$$

Equation (47) corresponds precisely with the classical jump condition across a discontinuous interface (Eringen and Maugin, 1990)

$$\mathbf{n} \cdot [[\mathcal{J} + q_e(\mathbf{v} - \mathbf{w})]] = 0. \quad (48)$$

4.6. LINEAR MOMENTUM BALANCE

The macroscale linear momentum equation is given by

$$\begin{aligned} \varepsilon^\alpha \rho^{\alpha j} \frac{D^{\alpha j} \mathbf{v}^{\alpha j}}{Dt} - \nabla \cdot (\varepsilon^\alpha \mathbf{t}^{\alpha j}) + \frac{1}{2} \varepsilon_0 \mathbf{E}_T \cdot \mathbf{E}^{\alpha j} \nabla \varepsilon^\alpha - \varepsilon^\alpha \rho^{\alpha j} (\mathbf{g}^{\alpha j} + \mathbf{g}_I^{\alpha j}) - \\ - \varepsilon^\alpha q_e^{\alpha j} \mathbf{E}_T - \varepsilon^\alpha \mathbf{P}^{\alpha j} \cdot \nabla \mathbf{E}_T = \varepsilon^\alpha \rho^{\alpha j} \widehat{\mathbf{i}}^{\alpha j} + \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^{\alpha j} \widehat{\mathbf{T}}_\beta^{\alpha j}, \end{aligned} \quad (49)$$

where $\mathbf{E}_T$ is the upscaled (total) electric field, $\mathbf{t}^{\alpha j}$ is the partial Cauchy stress tensor, $\mathbf{g}^{\alpha j}$ is the external body force acting on constituent j in phase α , and

$$\begin{aligned} \widehat{\mathbf{T}}_\beta^{\alpha j} = \frac{1}{\rho^{\alpha j} |\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} \left[(\mathbf{t}^j)^T + \mathbf{E} \mathbf{D}^j - \frac{1}{2} \varepsilon_0 \mathbf{E}^j \cdot \mathbf{E} \mathbf{I} + \right. \\ \left. + \rho^j \mathbf{v}^j (\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j) \right] \cdot \mathbf{n}^\alpha da - \frac{\mathbf{v}^{\alpha j}}{\rho^{\alpha j} |\delta V_\alpha|} \times \\ \times \int_{\delta A_{\alpha\beta}} \rho^j (\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j) \cdot \mathbf{n}^\alpha da + \frac{1}{\rho^{\alpha j}} \widehat{\mathbf{d}}_\beta^{\alpha j} \mathbf{E}_T, \end{aligned} \quad (50)$$

represents the effect constituent j of phase β has on the rate of change of mechanical momentum of the same constituent in phase α . The exchange term, $\widehat{\mathbf{i}}^{\alpha j}$, takes into account all gain of momenta due to the presence of other species but not due to chemical reactions. The relation between the other macroscale variables and their microscale counterparts are given in Appendix C. The last term on the left-hand side is usually referred to as the Kelvin force, and the adjacent term (on the LHS of Eq. (49)) is referred to as the Lorentz force (Eringen and Maugin, 1990). Note that unlike (Kelly, 1964; Eringen, 1998) we assume it is the *total* electric field which contributes to these forces.

There are a couple of additional forces in this form, when compared to the classical form due to the introduction of the species electric field, $\mathbf{E}^{\alpha j}$, and the volume fraction ε^α . The first we consider is an additional (internal) body force, $\mathbf{g}_I$. It is due to the difference between $\mathbf{E}$ and $\mathbf{E}^j$ (see Appendix C), and so the body force can no longer be considered due to gravity alone. The other additional term involves the gradient of the volume fraction. If the medium is homogeneous in volume fraction (ε^α is constant), then this term is zero. The form of this equation is motivated by the form of $\widehat{\mathbf{T}}_\beta^{\alpha j}$, which we choose to be consistent with the jump condition associated with momentum balance.

Summing over j yields

$$\begin{aligned} \varepsilon^\alpha \rho^\alpha \frac{D^\alpha \mathbf{v}^\alpha}{Dt} - \nabla \cdot (\varepsilon^\alpha \mathbf{t}^\alpha) - \varepsilon^\alpha \rho^\alpha (\mathbf{g}^\alpha + \mathbf{g}_I^\alpha) - \varepsilon^\alpha q_e^\alpha \mathbf{E}_T - \varepsilon^\alpha \mathbf{P}^\alpha \cdot \nabla \mathbf{E}_T + \\ + \frac{1}{2} \varepsilon_0 \mathbf{E}_T \cdot \mathbf{E}^\alpha \nabla \varepsilon^\alpha = \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^\alpha \widehat{\mathbf{T}}_\beta^\alpha, \end{aligned} \quad (51)$$

with restrictions

$$\sum_{j=1}^N \rho^{\alpha_j} (\widehat{\mathbf{i}}^{\alpha_j} + \widehat{\mathbf{r}}^{\alpha_j} \mathbf{v}^{\alpha_j}) = 0, \quad \forall \alpha, \quad (52)$$

$$\varepsilon^\alpha \rho^{\alpha_j} (\widehat{\mathbf{T}}_\beta^{\alpha_j} + \widehat{\mathbf{e}}_\beta^{\alpha_j} \mathbf{v}^{\alpha_j}) + \varepsilon^\beta \rho^{\beta_j} (\widehat{\mathbf{T}}_\alpha^{\beta_j} + \widehat{\mathbf{e}}_\alpha^{\beta_j} \mathbf{v}^{\beta_j}) = 0, \quad j = 1, \dots, N. \quad (53)$$

The jump condition across a discontinuous interface is (Eringen and Maugin, 1990)

$$\mathbf{n} \cdot [[\rho \mathbf{v}(\mathbf{v} - \mathbf{w}) - \mathbf{t} - \mathbf{t}_E]] = 0, \quad (54)$$

where $\mathbf{w}$ is the speed of the discontinuity and for the electroquasistatic case considered here,

$$\mathbf{F}_e = \nabla \cdot \mathbf{t}_E = \nabla \cdot \left(\mathbf{D}\mathbf{E} - \frac{1}{2} \varepsilon_0 \mathbf{E} \cdot \mathbf{E} \mathbf{I} \right). \quad (55)$$

Equation (54) can be shown to correspond directly with (53) by using (26).

4.7. ANGULAR MOMENTUM BALANCE

The macroscale form of the conservation of angular momentum equation in indicial notation is

$$\begin{aligned} -\varepsilon^\alpha \epsilon_{klm} \dot{\mathbf{t}}_{kl}^{\alpha_j} - \varepsilon^\alpha \epsilon_{klm} P_k^{\alpha_j} (E_T)_l \\ = \varepsilon^\alpha \rho^{\alpha_j} \widehat{\mathbf{m}}_m^{\alpha_j} + M_m^{\alpha_j} + \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^{\alpha_j} (\widehat{\mathbf{m}}_\beta^{\alpha_j})_m, \end{aligned} \quad (56)$$

where the definitions of all variables are given in Appendix C. Thus, the macroscale equation is composed of two parts. The equation consisting of all but the term involving $\mathbf{M}^{\alpha_j}$ is the macroscale form of the microscale equation. The term $\mathbf{M}^{\alpha_j}$ contains all terms arising from the microscale conservation of linear momentum crossed with $\boldsymbol{\xi}$ and volume averaged. These additional terms for the purely mechanical case have been derived before (Eringen and Suhubi, 1964; Hassanizadeh and Gray, 1979b). The term in $\mathbf{M}^{\alpha_j}$ involving the time rate of change of velocity is known as inertial (Eringen and Suhubi, 1964) or local (Hassanizadeh and Gray, 1979b) spin. The term in $\mathbf{M}^{\alpha_j}$ involving $\mathbf{t}^j$ has been referred to as the first surface stress moment (Eringen and Suhubi, 1964), or apparent couple stress tensor

(Hassanizadeh and Gray, 1979b). One observation which is painfully clear is that even in a medium which contains no electric field or charge, the mixture of two media which have symmetric stress tensors may result in a macroscopic medium which has a nonsymmetric stress tensor.

Summing over j yields

$$-\varepsilon^\alpha \epsilon_{klm} t_{kl}^\alpha - \varepsilon^\alpha \epsilon_{klm} P_k^\alpha (E_T)_l = M_m^\alpha + \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^\alpha (\hat{\mathbf{m}}_\beta^\alpha)_m, \quad (57)$$

where we note that symmetry of the stress tensor is lost due to possible polarization of the medium, the microscale angular momentum, and the interaction of the phase with other phases.

The restriction which must hold to ensure that the bulk-phase angular momentum can only be lost due to transfer to other phases is

$$\sum_{j=1}^N \rho^{\alpha_j} \hat{\mathbf{m}}^{\alpha_j} = \mathbf{0}, \quad \forall \alpha. \quad (58)$$

The restriction which arises from the assumption of no interfacial angular momentum is

$$\varepsilon^\alpha \rho^{\alpha_j} \hat{\mathbf{m}}_\beta^{\alpha_j} + \varepsilon^\beta \rho^{\beta_j} \hat{\mathbf{m}}_\alpha^{\beta_j} = 0, \quad j = 1, \dots, N. \quad (59)$$

4.8. CONSERVATION OF ENERGY

The conservation of energy equation is given by

$$\begin{aligned} & \varepsilon^\alpha \rho^{\alpha_j} \frac{D^{\alpha_j} e^{\alpha_j}}{Dt} - \varepsilon^\alpha \mathbf{t}^{\alpha_j} : \nabla \mathbf{v}^{\alpha_j} - \nabla \cdot (\varepsilon^\alpha \mathbf{q}^{\alpha_j}) - \frac{1}{2} \varepsilon_0 \mathbf{E}_T \cdot \mathbf{E}^{\alpha_j} \frac{D^{\alpha_j} \varepsilon^\alpha}{Dt} - \\ & - \varepsilon^\alpha \rho^{\alpha_j} h^{\alpha_j} - \varepsilon^\alpha \mathcal{J}^{\alpha_j} \cdot \mathbf{E}_T - \frac{\partial(\varepsilon^\alpha \mathbf{P}^{\alpha_j})}{\partial t} \cdot \mathbf{E}_T - \nabla \cdot (\varepsilon^\alpha \mathbf{v}^{\alpha_j} \mathbf{P}^{\alpha_j}) \cdot \mathbf{E}_T \\ & = \varepsilon^\alpha \rho^{\alpha_j} \hat{Q}^{\alpha_j} + \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^{\alpha_j} \hat{Q}_\beta^{\alpha_j}, \end{aligned} \quad (60)$$

where the colon indicates a tensor dot product (e.g. $\underline{\underline{a}}:\underline{\underline{b}} = \sum_{i,j} a_{ij} b_{ij}$), $\hat{Q}^{\alpha_j}$ represents the rate of energy gain due to the presence of other constituents in the same phase, and $\hat{Q}_\beta^{\alpha_j}$, given in Appendix C, represents the effect constituent j of phase β has on the rate of change of energy of the same constituent in phase α due to non-mass transfer, non-mechanical means.

The sum of the last three terms on the left-hand side is the electrical energy source. As in the linear momentum balance we assume that electric energy is due to the *total* electric field, not just part of it. These three terms take on a variety of forms (Melcher, 1981; Eringen and Maugin, 1990), which are related through the Maxwell equations (Gauss' law, Faraday's law, Ampère's law, and Conservation

of Electric Charge). The form presented here is the one derived directly from microscale electrical forces using a statistical averaging approach (see Eringen and Maugin, 1990) making no use of Maxwell's equations, as otherwise exchange terms $(\widehat{d}^j, \widehat{\sigma}^j, \widehat{q}^j)$ appear.

As in the linear momentum equation, upscaling produces an additional term involving the volume fraction, but in this case, the material time rate of change of the volume fraction. This term would be negligible in a non-swelling porous medium. Additional effects of fluctuations in the species electric work terms manifest themselves in the flux term, $\mathbf{q}^{\alpha j}$, and the external source term, $h^{\alpha j}$.

Summing over constituents yields

$$\begin{aligned} & \varepsilon^\alpha \rho^\alpha \frac{D^\alpha e^\alpha}{Dt} - \varepsilon^\alpha \mathbf{t}^\alpha : \nabla \mathbf{v}^\alpha - \nabla \cdot (\varepsilon^\alpha \mathbf{q}^\alpha) - \varepsilon^\alpha \rho^\alpha h^\alpha - \varepsilon^\alpha \mathcal{J}^\alpha \cdot \mathbf{E}_T - \\ & - \frac{\partial(\varepsilon^\alpha \mathbf{P}^\alpha)}{\partial t} \cdot \mathbf{E}_T - \nabla \cdot (\varepsilon^\alpha \mathbf{v}^\alpha \mathbf{P}^\alpha) \cdot \mathbf{E}_T - \frac{1}{2} \varepsilon_0 \mathbf{E}^\alpha \cdot \mathbf{E}_T \frac{D^\alpha \varepsilon^\alpha}{Dt} \\ & = \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^\alpha \widehat{Q}_\beta^\alpha, \end{aligned} \quad (61)$$

with the restrictions

$$\sum_{j=1}^N \left[\rho^{\alpha j} \widehat{Q}^{\alpha j} + \rho^{\alpha j} \widehat{\mathbf{i}}^{\alpha j} \cdot \mathbf{v}^{\alpha j} + \rho^{\alpha j} \widehat{\mathbf{r}}^{\alpha j} \left(e^{\alpha j} + \frac{1}{2} (\mathbf{v}^{\alpha j})^2 \right) \right] = 0, \quad \forall \alpha, \quad (62)$$

$$\begin{aligned} & \left[\varepsilon^\alpha \rho^{\alpha j} \widehat{Q}_\beta^{\alpha j} + \varepsilon^\alpha \rho^{\alpha j} \widehat{\mathbf{T}}_\beta^{\alpha j} \cdot \mathbf{v}^{\alpha j} + \varepsilon^\alpha \rho^{\alpha j} \widehat{\mathbf{e}}_\beta^{\alpha j} \left(e^{\alpha j} + \frac{1}{2} (\mathbf{v}^{\alpha j})^2 \right) \right] + \\ & + \left[\varepsilon^\beta \rho^{\beta j} \widehat{Q}_\alpha^{\beta j} + \varepsilon^\beta \rho^{\beta j} \widehat{\mathbf{T}}_\alpha^{\beta j} \cdot \mathbf{v}^{\beta j} + \right. \\ & \left. + \varepsilon^\beta \rho^{\beta j} \widehat{\mathbf{e}}_\alpha^{\beta j} \left(e^{\beta j} + \frac{1}{2} (\mathbf{v}^{\beta j})^2 \right) \right] = 0, \quad j = 1, \dots, N. \end{aligned} \quad (63)$$

The first restriction states that bulk-phase energy can only be lost due to transfer to other phases. The second states that the interface can hold no energy.

We wish to compare (63) with the jump discontinuity condition. Various forms exist depending on what is considered negligible and the use of Maxwell's equations. Here, we consider

$$\begin{aligned} & [[\mathbf{q} + \mathbf{t} \cdot \mathbf{v} + \left(\rho e + \frac{1}{2} \rho v^2 \right) (\mathbf{w} - \mathbf{v}) + \\ & + \frac{1}{2} \varepsilon_0 (\mathbf{E} \cdot \mathbf{E}) \mathbf{w} + \mathbf{H} \times \mathbf{E} + \mathbf{P} \mathbf{E} \cdot \mathbf{v}]] \cdot \mathbf{n} = 0. \end{aligned} \quad (64)$$

Equation (63) can be re-written as

$$\begin{aligned}
& \frac{1}{|\delta V|} \int_{\delta A_{\alpha\beta}} \left[\mathbf{q}^j + (\mathbf{t}^j + \mathbf{P}^j \mathbf{E}) \cdot \mathbf{v}^j + \rho^j \left(e^j + \frac{1}{2} (\mathbf{v}^j)^2 \right) (\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j) + \right. \\
& \quad \left. + \frac{1}{2} \varepsilon_0 (\mathbf{E}^j \cdot \mathbf{E}) \mathbf{w}_{\alpha\beta}^j + \mathbf{H}^j \times \mathbf{E} \right] \cdot \mathbf{n}^\alpha \, da \\
& = \frac{1}{|\delta V|} \int_{\delta A_{\alpha\beta}} \left[\mathbf{q}^j + (\mathbf{t}^j + \mathbf{P}^j \mathbf{E}) \cdot \mathbf{v}^j + \rho^j \left(e^j + \frac{1}{2} (\mathbf{v}^j)^2 \right) (\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j) + \right. \\
& \quad \left. + \frac{1}{2} \varepsilon_0 (\mathbf{E}^j \cdot \mathbf{E}) \mathbf{w}_{\alpha\beta}^j + \mathbf{H}^j \times \mathbf{E} \right] \cdot \mathbf{n}^\beta \, da.
\end{aligned} \tag{65}$$

Comparing terms we see that the condition involving exchange of energy across the interface, (63), conforms exactly with the classical jump condition (64).

4.9. ENTROPY BALANCE

The entropy balance for constituent j in phase α is given by

$$\begin{aligned}
& \varepsilon^\alpha \rho^{\alpha j} \frac{D^{\alpha j} \eta^{\alpha j}}{Dt} - \nabla \cdot \varepsilon^\alpha \boldsymbol{\phi}^{\alpha j} - \varepsilon^\alpha \rho^{\alpha j} b^{\alpha j} \\
& = \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^{\alpha j} \widehat{\Phi}_\beta^{\alpha j} + \varepsilon^\alpha \rho^{\alpha j} \widehat{\eta}^{\alpha j} + \varepsilon^\alpha \rho^{\alpha j} \widehat{\Lambda}^{\alpha j},
\end{aligned} \tag{66}$$

where $\widehat{\Phi}_\beta^{\alpha j}$ represents the rate at which entropy is gained from constituent j in phase β . Summing over constituents, we get

$$\varepsilon^\alpha \rho^\alpha \frac{D^\alpha \eta^\alpha}{Dt} - \nabla \cdot (\varepsilon^\alpha \boldsymbol{\phi}^\alpha) - \varepsilon^\alpha \rho^\alpha b^\alpha = \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^\alpha \widehat{\Phi}_\beta^\alpha + \varepsilon^\alpha \rho^\alpha \widehat{\Lambda}^\alpha, \tag{67}$$

with restrictions

$$\sum_{j=1}^N \rho^{\alpha j} (\widehat{\eta}^{\alpha j} + \widehat{r}^{\alpha j} \eta^{\alpha j}) = 0, \quad \forall \alpha. \tag{68}$$

$$\varepsilon^\alpha \rho^{\alpha j} (\widehat{\Phi}_\beta^{\alpha j} + \widehat{e}_\beta^{\alpha j} \eta^{\alpha j}) + \varepsilon^\beta \rho^{\beta j} (\widehat{\Phi}_\alpha^{\beta j} + \widehat{e}_\alpha^{\beta j} \eta^{\beta j}) = 0,$$

$j = 1, \dots, N. \tag{69}$

The second condition is exactly analogous to the classical jump boundary condition applied at an interface. This equation can be used to obtain restrictions on the forms of constitutive equations using, for example, the Coleman and Noll method (see e.g. Eringen (1967), Bennethum and Cushman (1996b)). At this point we make no assumptions regarding the form of the entropy flux or source. It is now postulated

that the rate of entropy production is non-negative after summing over both species and phases.

5. Discussion

We have provided the macroscale balance laws for a multiphase, multi-species, swelling system, which incorporates electroquasistatics. The microscale equations from which these equations are derived follow the approach of Eringen and Maugin (1990), while the upscaling to the macroscale is achieved via spatial averaging. The species electric field, $\mathbf{E}^{\alpha_j}$, is carefully defined and the complete relations between the microscopic and macroscopic variables and the species and bulk phase variables are provided. Further comparisons were made between restrictions required to hold at interfaces and jump conditions classically used at interfaces.

In regards to momentum balance and conservation of energy, there are several ways in which one can group together terms into the definitions of the macroscopic variables. In this paper, we choose the macroscopic definitions in a way which results in consistency between restriction conditions at interfaces and classical jump conditions used at interfaces. But as a consequence, additional body sources appear due not to external sources but due to species electric fields and microscopic fluctuations of the work terms. Further, there are additional terms which appear due to gradients in volume fractions and material time rate-of-change of the volume fractions in the balance of momentum and conservation of energy equation, respectively. This is *not* due to the primitive choice of $\mathbf{F}_e^j$ and W_e^j (see Equations (2)–(4)). Rather it is a consequence of the way terms are grouped so as to recover classical jump conditions. The consequence of these additional terms needs to be investigated. This is initially explored in Part II of these papers.

The field equations are required for many practical applications including, but not limited to, swelling clays (smectites), biopolymers, biological membranes, pulsed electrophoresis, and chromatography. In Part II of these papers we exploit the entropy inequality using independent variables consistent with several natural systems and apply the resulting constitutive theory near equilibrium for two problems: electrolyte transport in swelling clays and pulsed electrophoresis.

Appendix A. Definition of Macroscopic Bulk Variables

The relationships between the macroscopic constituent variables and their bulk counterparts follow:

$$\rho^\alpha = \sum_{j=1}^N \rho^{\alpha_j}, \quad (\text{B.1})$$

$$C^{\alpha_j} = \frac{\rho^{\alpha_j}}{\rho^\alpha}, \quad (\text{B.2})$$

$$\mathbf{v}^\alpha = \sum_{j=1}^N C^{\alpha_j} \mathbf{v}^{\alpha_j}, \quad (\text{B.3})$$

$$\widehat{\mathbf{e}}_\beta^\alpha = \sum_{j=1}^N C^{\alpha_j} \widehat{\mathbf{e}}_\beta^{\alpha_j}, \quad (\text{B.4})$$

$$\mathbf{J}^\alpha = \sum_{j=1}^N \mathbf{J}^{\alpha_j}, \quad (\text{B.5})$$

$$\mathcal{J}^\alpha = \sum_{j=1}^N \mathcal{J}^{\alpha_j} + q_e^{\alpha_j} \mathbf{v}^{\alpha_j, \alpha}, \quad (\text{B.6})$$

$$\mathbf{D}^\alpha = \sum_{j=1}^N \mathbf{D}^{\alpha_j}, \quad (\text{B.7})$$

$$\mathbf{P}^\alpha = \sum_{j=1}^N \mathbf{P}^{\alpha_j}, \quad (\text{B.8})$$

$$\mathbf{E}^\alpha = \sum_{j=1}^N \mathbf{E}^{\alpha_j}, \quad (\text{B.9})$$

$$q_e^\alpha = \sum_{j=1}^N q_e^{\alpha_j}, \quad (\text{B.10})$$

$$\widehat{\mathbf{d}}_\beta^\alpha = \sum_{j=1}^N \widehat{\mathbf{d}}_\beta^{\alpha_j}, \quad (\text{B.11})$$

$$\widehat{\boldsymbol{\sigma}}_\beta^\alpha = \sum_{j=1}^N \widehat{\boldsymbol{\sigma}}_\beta^{\alpha_j}, \quad (\text{B.12})$$

$$\mathbf{H}^\alpha = \sum_{j=1}^N (\mathbf{H}^{\alpha_j} - \mathbf{P}^{\alpha_j} \times \mathbf{v}^{\alpha_j, \alpha}), \quad (\text{B.13})$$

$$\widehat{\mathbf{h}}_\beta^\alpha = \sum_{j=1}^N \widehat{\mathbf{h}}_\beta^{\alpha_j}, \quad (\text{B.14})$$

$$z^\alpha = \sum_{j=1}^N C^{\alpha_j} z^{\alpha_j}, \quad (\text{B.15})$$

$$\widehat{Z}_\beta^\alpha = \sum_{j=1}^N C^{\alpha_j} (\widehat{Z}_\beta^{\alpha_j} + \widehat{e}_\beta^{\alpha_j} z^{\alpha_j, \alpha}), \quad (\text{B.16})$$

$$\mathbf{t}^\alpha = \sum_{j=1}^N [\mathbf{t}^{\alpha_j} - \rho^{\alpha_j} \mathbf{v}^{\alpha_j, \alpha} \mathbf{v}^{\alpha_j, \alpha}], \quad (\text{B.17})$$

$$\mathbf{g}^\alpha = \sum_{j=1}^N C^{\alpha_j} \mathbf{g}^{\alpha_j}, \quad (\text{B.18})$$

$$\mathbf{g}_I^\alpha = \sum_{j=1}^N C^{\alpha_j} \mathbf{g}_I^{\alpha_j}, \quad (\text{B.19})$$

$$\widehat{\mathbf{T}}_\beta^\alpha = \sum_{j=1}^N C^{\alpha_j} (\widehat{\mathbf{T}}_\beta^{\alpha_j} + \widehat{e}_\beta^{\alpha_j} \mathbf{v}^{\alpha_j, \alpha}) + \frac{1}{\rho^\alpha} \widehat{d}_\beta^\alpha \mathbf{E}_T, \quad (\text{B.20})$$

$$\widehat{\mathbf{m}}_\beta^\alpha = \sum_{j=1}^N C^{\alpha_j} \widehat{\mathbf{m}}_\beta^{\alpha_j}, \quad (\text{B.21})$$

$$\mathbf{M}^\alpha = \sum_{j=1}^N \mathbf{M}^{\alpha_j}, \quad (\text{B.22})$$

$$e^\alpha = \sum_{j=1}^N C^{\alpha_j} \left(e^{\alpha_j} + \frac{1}{2} \mathbf{v}^{\alpha_j, \alpha} \cdot \mathbf{v}^{\alpha_j, \alpha} \right), \quad (\text{B.23})$$

$$\begin{aligned} \mathbf{q}^\alpha = \sum_{j=1}^N & \left[\mathbf{q}^{\alpha_j} + \mathbf{t}^{\alpha_j} \cdot \mathbf{v}^{\alpha_j, \alpha} - \right. \\ & \left. - \rho^{\alpha_j} \mathbf{v}^{\alpha_j, \alpha} \left(e^{\alpha_j} + \frac{1}{2} (\mathbf{v}^{\alpha_j, \alpha})^2 \right) + \mathbf{v}^{\alpha_j, \alpha} \mathbf{P}^{\alpha_j} \cdot \mathbf{E}_T \right], \end{aligned} \quad (\text{B.24})$$

$$h^\alpha = \sum_{j=1}^N C^{\alpha_j} (h^{\alpha_j} + (\mathbf{g}^{\alpha_j} + \mathbf{g}_I^{\alpha_j}) \cdot \mathbf{v}^{\alpha_j, \alpha}), \quad (\text{B.25})$$

$$\begin{aligned}\widehat{Q}_\beta^\alpha = & \sum_{j=1}^N C^{\alpha_j} \left[\widehat{Q}_\beta^{\alpha_j} + \widehat{\mathbf{T}}_\beta^{\alpha_j} \cdot \mathbf{v}^{\alpha_j, \alpha} + \widehat{e}_\beta^{\alpha_j} \left(e^{\alpha_j, \alpha} + \frac{1}{2} (\mathbf{v}^{\alpha_j, \alpha})^2 \right) \right] - \\ & - \frac{1}{\rho^\alpha} [\widehat{d}_\beta^\alpha \mathbf{E}_t \cdot \mathbf{v}^\alpha + \widehat{\mathbf{h}}_\beta^\alpha \cdot \mathbf{E}_T],\end{aligned}\quad (\text{B.26})$$

$$\eta^\alpha = \sum_{j=1}^N C^{\alpha_j} \eta^{\alpha_j}, \quad (\text{B.27})$$

$$\phi^\alpha = \sum_{j=1}^N (\phi^{\alpha_j} - \rho^{\alpha_j} \eta^{\alpha_j} \mathbf{v}^{\alpha_j, \alpha}), \quad (\text{B.28})$$

$$b^\alpha = \sum_{j=1}^N C^{\alpha_j} b^{\alpha_j}, \quad (\text{B.29})$$

$$\widehat{\Phi}_\beta^\alpha = \sum_{j=1}^N C^{\alpha_j} (\widehat{\Phi}_\beta^{\alpha_j} + \widehat{e}_\beta^{\alpha_j} \eta^{\alpha_j, \alpha}), \quad (\text{B.30})$$

$$\widehat{\Lambda}^\alpha = \sum_{j=1}^N C^{\alpha_j} \widehat{\Lambda}^{\alpha_j}. \quad (\text{B.31})$$

Appendix B. Relationship between Macroscopic and Microscopic Variables

For notational brevity, define the volume average over phase α with angle brackets:

$$\langle \Psi^j \rangle^\alpha(\mathbf{x}, t) = \frac{1}{|\delta V_\alpha|} \int_{\delta V} \Psi^j(\mathbf{r}, t) \gamma_\alpha(\mathbf{r}, t) dv(\boldsymbol{\xi}) \quad (\text{C.1})$$

so that if for example, ρ^j is constant throughout phase α then $\langle \rho^j \rangle^\alpha$ would be that constant density.

Similarly we define the mass average to be

$$\overline{\Psi^j}^\alpha(\mathbf{x}, t) = \frac{1}{\langle \rho^j \rangle^\alpha |\delta V_\alpha|} \int_{\delta V} \rho^j(\mathbf{r}, t) \Psi^j(\mathbf{r}, t) \gamma_\alpha(\mathbf{r}, t) dv(\boldsymbol{\xi}). \quad (\text{C.2})$$

Thus for example, while the volume average of velocity makes no physical sense, the mass average (momentum) does.

The relationships between the macroscopic constituent variables and their microscopic counterparts follow. There are *no* assumptions made about small fluctuations about the average. What follows is exact.

$$\rho^{\alpha_j} = \langle \rho^j \rangle^\alpha, \quad (\text{C.3})$$

$$\mathbf{v}^{\alpha_j} = \overline{\mathbf{v}^j}^\alpha, \quad (\text{C.4})$$

$$\widehat{\mathbf{r}}^{\alpha_j} = \widehat{\mathbf{r}^j}^\alpha, \quad (\text{C.5})$$

$$\widehat{\mathbf{e}}_\beta^{\alpha_j} = \frac{1}{\rho^{\alpha_j} |\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} \rho^j (\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j) \cdot \mathbf{n}^\alpha \, da, \quad (\text{C.6})$$

$$\mathbf{J}^{\alpha_j} = \langle \mathbf{J}^j \rangle^\alpha, \quad (\text{C.7})$$

$$\mathcal{J}^{\alpha_j} = \langle \mathcal{J}^j \rangle^\alpha + \langle q_e^j \mathbf{v}^j \rangle^\alpha - q_e^{\alpha_j} \mathbf{v}^{\alpha_j}, \quad (\text{C.8})$$

$$\mathbf{D}^{\alpha_j} = \langle \mathbf{D}^j \rangle^\alpha, \quad (\text{C.9})$$

$$\mathbf{P}^{\alpha_j} = \langle \mathbf{P}^j \rangle^\alpha, \quad (\text{C.10})$$

$$\mathbf{E}^{\alpha_j} = \langle \mathbf{E}^j \rangle^\alpha, \quad (\text{C.11})$$

$$\mathbf{E}_T = \sum_\alpha \varepsilon^\alpha \langle \mathbf{E} \rangle^\alpha = \frac{1}{|\delta V|} \int_{\delta V} \mathbf{E} \, dv, \quad (\text{C.12})$$

$$q_e^{\alpha_j} = \langle q_e^j \rangle^\alpha, \quad (\text{C.13})$$

$$\widehat{\mathbf{d}}^{\alpha_j} = \langle \widehat{\mathbf{d}}^j \rangle^\alpha, \quad (\text{C.14})$$

$$\widehat{\mathbf{d}}_\beta^{\alpha_j} = -\frac{1}{|\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} \mathbf{D}^j \cdot \mathbf{n}^\alpha \, da, \quad (\text{C.15})$$

$$\widehat{\boldsymbol{\sigma}}^{\alpha_j} = \langle \widehat{\boldsymbol{\sigma}}^j \rangle^\alpha, \quad (\text{C.16})$$

$$\widehat{\boldsymbol{\sigma}}_\beta^{\alpha_j} = -\frac{1}{|\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} \mathbf{n}^\alpha \times \mathbf{E}^j \, da, \quad (\text{C.17})$$

$$\mathbf{H}^{\alpha_j} = \langle \mathbf{H}^j \rangle^\alpha + [\mathbf{P}^{\alpha_j} \times \mathbf{v}^{\alpha_j} - \langle \mathbf{P}^j \times \mathbf{v}^j \rangle^\alpha], \quad (\text{C.18})$$

$$\widehat{\mathbf{h}}_\beta^{\alpha_j} = \frac{1}{|\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} [\mathbf{D}^j \mathbf{w}_{\alpha\beta}^j \cdot \mathbf{n}^\alpha - (\mathbf{H}^j - \mathbf{P}^j \times \mathbf{v}^j) \times \mathbf{n}^\alpha] \, da, \quad (\text{C.19})$$

$$\widehat{\mathbf{h}}^{\alpha_j} = \langle \widehat{\mathbf{h}}^j \rangle^\alpha, \quad (\text{C.20})$$

$$\mathbf{z}^{\alpha_j} = \overline{\mathbf{z}^j}, \quad (\text{C.21})$$

$$\widehat{\mathbf{q}}^{\alpha_j} = \langle \widehat{\mathbf{q}}^j \rangle^\alpha + \overline{\mathbf{z}^j \widehat{\mathbf{r}}^j}^\alpha - \widehat{\mathbf{r}}^{\alpha_j} \mathbf{z}^{\alpha_j}, \quad (\text{C.22})$$

$$\begin{aligned}\widehat{Z}_\beta^{\alpha_j} &= \frac{1}{\rho^{\alpha_j} |\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} [q_e^j(\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j) - \mathcal{J}^j] \cdot \mathbf{n}^\alpha \, da - \\ &\quad - \frac{\widehat{z}^{\alpha_j}}{\rho^{\alpha_j} |\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} [\rho^j(\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j)] \cdot \mathbf{n}^\alpha \, da,\end{aligned}\quad (\text{C.23})$$

$$\begin{aligned}\mathbf{t}^{\alpha_j} &= \langle \mathbf{t}^j \rangle^\alpha + \rho^{\alpha_j} \mathbf{v}^{\alpha_j} \mathbf{v}^{\alpha_j} - \rho^{\alpha_j} \overline{\mathbf{v}^j \mathbf{v}^j}^\alpha + \langle \mathbf{D}^j \mathbf{E} \rangle^\alpha - \mathbf{D}^{\alpha_j} \mathbf{E}_T + \\ &\quad + \frac{1}{2} \varepsilon_0 [\mathbf{E}^{\alpha_j} \cdot \mathbf{E}_T - \langle \mathbf{E}^j \cdot \mathbf{E} \rangle^\alpha] \mathbf{I},\end{aligned}\quad (\text{C.24})$$

$$\mathbf{g}^{\alpha_j} = \overline{\mathbf{g}^j}^\alpha, \quad (\text{C.25})$$

$$\begin{aligned}\mathbf{g}_I^{\alpha_j} &= \frac{\varepsilon_0}{2\rho^{\alpha_j}} [\nabla \mathbf{E}_T \cdot \mathbf{E}^{\alpha_j} - \langle \nabla \mathbf{E} \cdot \mathbf{E}^j \rangle^\alpha - \\ &\quad - \nabla \mathbf{E}^{\alpha_j} \cdot \mathbf{E}_T + \langle \nabla \mathbf{E}^j \cdot \mathbf{E} \rangle^\alpha],\end{aligned}\quad (\text{C.26})$$

$$\widehat{\mathbf{i}}^{\alpha_j} = \widehat{\mathbf{i}}^j{}^\alpha + \widehat{\mathbf{r}}^j \mathbf{v}^j{}^\alpha - \widehat{\mathbf{r}}^{\alpha_j} \mathbf{v}^{\alpha_j} + \frac{1}{\rho^{\alpha_j}} (\widehat{d}^{\alpha_j} \mathbf{E}_T - \langle \widehat{d}^j \mathbf{E} \rangle^\alpha), \quad (\text{C.27})$$

$$\begin{aligned}\widehat{\mathbf{T}}_\beta^{\alpha_j} &= \frac{1}{\rho^{\alpha_j} |\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} \times \left[(\mathbf{t}^j)^T + \mathbf{E} \mathbf{D}^j - \frac{1}{2} \varepsilon_0 \mathbf{E}^j \cdot \mathbf{E} \mathbf{I} + \right. \\ &\quad \left. + \rho^j \mathbf{v}^j (\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j) \right] \cdot \mathbf{n}^\alpha \, da - \\ &\quad - \frac{\mathbf{v}^{\alpha_j}}{\rho^{\alpha_j} |\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} \rho^j (\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j) \cdot \mathbf{n}^\alpha \, da + \frac{1}{\rho^{\alpha_j}} \widehat{d}_\beta^{\alpha_j} \mathbf{E}_T,\end{aligned}\quad (\text{C.28})$$

$$\widehat{\mathbf{m}}^{\alpha_j} = \langle \widehat{\mathbf{m}}^j \rangle^\alpha - \frac{1}{\rho^j} [\langle \mathbf{P}^j \times \mathbf{E}^j \rangle^\alpha - \mathbf{P}^{\alpha_j} \times \mathbf{E}^{\alpha_j}], \quad (\text{C.29})$$

$$(\widehat{\mathbf{m}}_\beta^{\alpha_j})_m = \frac{1}{\rho^{\alpha_j} |\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} [\epsilon_{klm} \xi_k (t_{nl} + v_l ((w_{\alpha\beta}^j)_n - v_n^j)) n_n] \, da, \quad (\text{C.30})$$

$$\begin{aligned}(M^{\alpha_j})_m &= -\frac{\partial}{\partial t} (\varepsilon^\alpha \rho^{\alpha_j} \overline{\epsilon_{klm} \xi_k v_l^j}^\alpha) - [\varepsilon^\alpha \rho^{\alpha_j} v_n^j (\overline{\epsilon_{klm} \xi_k v_l^j}^\alpha)]_{,n} + \\ &\quad + [\varepsilon^\alpha \langle \epsilon_{klm} \xi_k \mathbf{t}_{nl}^j \rangle^\alpha]_{,n} + \\ &\quad + \varepsilon^\alpha \rho^{\alpha_j} \overline{\epsilon_{klm} \xi_k g_l^j}^\alpha + \varepsilon^\alpha \langle \epsilon_{klm} \xi_k (F_e^j)_l \rangle^\alpha + \\ &\quad + \varepsilon^\alpha \langle \epsilon_{klm} \xi_k (\rho^j \widehat{\mathbf{i}}_k^j + \rho^j \widehat{\mathbf{r}}^j v_k^j) \rangle^\alpha,\end{aligned}\quad (\text{C.31})$$

$$\begin{aligned}
e^{\alpha_j} &= \overline{e^j}^\alpha + \frac{1}{2} \overline{\mathbf{v}^j \cdot \mathbf{v}^j}^\alpha - \\
&\quad - \frac{1}{2} \mathbf{v}^{\alpha_j} \cdot \mathbf{v}^{\alpha_j} + \frac{\varepsilon_0}{\rho^{\alpha_j}} [\langle \mathbf{E}^j \cdot \mathbf{E} \rangle^\alpha - \mathbf{E}^{\alpha_j} \cdot \mathbf{E}_T],
\end{aligned} \tag{C.32}$$

$$\begin{aligned}
\mathbf{q}^{\alpha_j} &= \langle \mathbf{q}^j \rangle^\alpha + \langle \mathbf{t}^j \cdot \mathbf{v}^j \rangle^\alpha - \mathbf{t}^{\alpha_j} \cdot \mathbf{v}^{\alpha_j} + \\
&\quad + \rho^{\alpha_j} \mathbf{v}^{\alpha_j} \left(e^{\alpha_j} + \frac{1}{2} \mathbf{v}^{\alpha_j} \cdot \mathbf{v}^{\alpha_j} \right) - \overline{\rho^{\alpha_j} \mathbf{v}^j \left(e^j + \frac{1}{2} \mathbf{v}^j \cdot \mathbf{v}^j \right)}^\alpha + \\
&\quad + \langle \mathbf{E} \times \mathbf{H}^j \rangle^\alpha - \mathbf{E}_T \times \mathbf{H}^{\alpha_j} + \\
&\quad + \langle \mathbf{P}^j \mathbf{v}^j \cdot \mathbf{E} \rangle^\alpha - \mathbf{P}^{\alpha_j} \mathbf{v}^{\alpha_j} \cdot \mathbf{E}_T,
\end{aligned} \tag{C.33}$$

$$\begin{aligned}
h^{\alpha_j} &= \overline{h^j}^\alpha + \overline{\mathbf{g}^j \cdot \mathbf{v}^j}^\alpha - \overline{\mathbf{g}^j}^\alpha \cdot \mathbf{v}^{\alpha_j} + \\
&\quad + \frac{\varepsilon_0}{2\rho^{\alpha_j}} \left[\frac{\partial \mathbf{E}^{\alpha_j}}{\partial t} \cdot \mathbf{E}_T - \left\langle \frac{\partial \mathbf{E}^j}{\partial t} \cdot \mathbf{E}_T \right\rangle^\alpha + \right. \\
&\quad \left. + \left\langle \frac{\partial \mathbf{E}}{\partial t} \cdot \mathbf{E}^j \right\rangle^\alpha - \frac{\partial \mathbf{E}_T}{\partial t} \cdot \mathbf{E}^{\alpha_j} \right],
\end{aligned} \tag{C.34}$$

$$\begin{aligned}
\widehat{Q}^{\alpha_j} &= \overline{\widehat{Q}^j}^\alpha + \overline{\widehat{\mathbf{i}}^j \cdot \mathbf{v}^j}^\alpha - \widehat{\mathbf{i}}^{\alpha_j} \cdot \mathbf{v}^{\alpha_j} + \\
&\quad + \overline{\left(e^j + \frac{1}{2} \mathbf{v}^j \cdot \mathbf{v}^j \right) \widehat{r}^j}^\alpha - \overline{\left(e^j + \frac{1}{2} \mathbf{v}^j \cdot \mathbf{v}^j \right) \widehat{r}^{\alpha_j}}^\alpha + \\
&\quad + \frac{1}{\rho^{\alpha_j}} [\langle \widehat{\mathbf{h}}^j \cdot \mathbf{E} \rangle^\alpha - \widehat{\mathbf{h}}^{\alpha_j} \cdot \mathbf{E}_T],
\end{aligned} \tag{C.35}$$

$$\begin{aligned}
\widehat{Q}_\beta^{\alpha_j} &= \frac{1}{\rho^{\alpha_j} |\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} \left[\mathbf{q}^j + \left(\mathbf{t}^j + \mathbf{D}^j \mathbf{E} - \frac{1}{2} \varepsilon_0 (\mathbf{E}^j \cdot \mathbf{E}) \mathbf{I} \right) \cdot \mathbf{v}^j + \right. \\
&\quad \left. + \rho^j \left(e^j + \frac{1}{2} \mathbf{v}^j \cdot \mathbf{v}^j + \frac{1}{2} \varepsilon_0 \mathbf{E}^j \cdot \mathbf{E} \right) (\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j) \right] \cdot \mathbf{n}^\alpha \, da - \\
&\quad - \frac{e^{\alpha_j} - \frac{1}{2} \mathbf{v}^{\alpha_j} \cdot \mathbf{v}^{\alpha_j}}{\rho^{\alpha_j} |\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} \rho^j (\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j) \cdot \mathbf{n}^\alpha \, da - \frac{\mathbf{v}^{\alpha_j}}{\rho^{\alpha_j} |\delta V_\alpha|} \times \\
&\quad \times \int_{\delta A_{\alpha\beta}} \left[(\mathbf{t}^j)^T + \mathbf{E} \mathbf{D}^j - \frac{1}{2} \varepsilon_0 (\mathbf{E}^j \cdot \mathbf{E}) \mathbf{I} + \rho^j \mathbf{v}^j (\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j) \right] \times \\
&\quad \times \mathbf{n}^\alpha \, da + \frac{1}{\rho^{\alpha_j} |\delta V_\alpha|} \int_{\delta A_{\alpha\beta}} [(-\varepsilon_0 \mathbf{E}^j \mathbf{E} + \varepsilon_0 (\mathbf{E}^j \cdot \mathbf{E}) \mathbf{I}) \cdot \mathbf{v}^j + \\
&\quad + \mathbf{H}^j \times \mathbf{E}] \cdot \mathbf{n}^\alpha \, da - \widehat{d}_\beta^{\alpha_j} \mathbf{E}_T \cdot \mathbf{v}^{\alpha_j} - \widehat{\mathbf{h}}_\beta^{\alpha_j} \cdot \mathbf{E}_T,
\end{aligned} \tag{C.36}$$

$$\eta^{\alpha_j} = \overline{\eta^j}^\alpha, \quad (C.37)$$

$$\phi^{\alpha_j} = \langle \phi^j \rangle^\alpha + \rho^{\alpha_j} \mathbf{v}^{\alpha_j} \eta^{\alpha_j} - \rho^{\alpha_j} \overline{\mathbf{v}^j \eta^j}^\alpha, \quad (C.38)$$

$$b^{\alpha_j} = \overline{b^j}^\alpha, \quad (C.39)$$

$$\begin{aligned} \widehat{\Phi}^{\alpha_j} = & \frac{1}{\rho^{\alpha_j} |\delta V \alpha|} \int_{\delta A_{\alpha\beta}} [\phi^j + \rho^j \eta^j (\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j)] \cdot \mathbf{n}^\alpha da - \\ & - \frac{\eta^{\alpha_j}}{\rho^{\alpha_j} |\delta V \alpha|} \int_{\delta A_{\alpha\beta}} \rho^j (\mathbf{w}_{\alpha\beta}^j - \mathbf{v}^j) \cdot \mathbf{n}^\alpha da, \end{aligned} \quad (C.40)$$

$$\widehat{\eta}^{\alpha_j} = \overline{\eta^j}^\alpha + \overline{\widehat{r}^j \eta^j}^\alpha - \widehat{r}^{\alpha_j} \eta^{\alpha_j}, \quad (C.41)$$

$$\widehat{\Lambda}^{\alpha_j} = \overline{\Lambda^j}^\alpha. \quad (C.42)$$

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